

Safe Manipulation in Dynamic Environments via Online Perception-Based Neural Control Barrier Functions

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Abstract—We study collision avoidance for a manipulator that shares its workspace with a second manipulator executing its own task. The second arm is a moving, articulated, non-convex obstacle: its joint configuration and velocity are communicated to the ego controller, but converting that state into a control-level safety constraint at every control step requires an explicit geometric model and repeated distance computations. We instead learn a neural control barrier function (CBF) on a ray-based observation of the obstacle arm. Our key observation is that, when the obstacle moves, the time derivative of the composed barrier contains an obstacle-motion term induced through the observation. Because ray casting is not differentiable at hit switches and miss events, we learn a smooth surrogate of the sensor map and differentiate the surrogate only to approximate the observation sensitivities required by the CBF condition. The neural barrier itself is evaluated on the real ray observation $o = O(q, q_e)$ during both training and online filtering, while the surrogate supplies approximations of $\partial O / \partial q$ and $\partial O / \partial q_e$. In a verified evaluation over 1000 randomized trials with two 7-DoF Franka Panda arms, the proposed filter raises the collision-free rate from 64.0% to 82.0% and the task-success rate from 86.8% to 96.8% relative to a closely matched static-environment neural-CBF baseline; both differences are significant under paired McNemar tests ($p < 0.001$). An offline derivative diagnostic further shows smaller post-hoc CBF residuals with automatic differentiation of the observation sensitivities through the smooth surrogate than with finite differences. We state the assumptions—communicated obstacle state, simulation-only evaluation, and a learned rather than formally verified barrier—explicitly.

I. INTRODUCTION

Manipulation cells increasingly contain more than one robot: dual-arm stations, a mobile manipulator that assists a fixed arm, or several arms serving one conveyor. Each arm follows its own task, so from the viewpoint of any one of them—the *ego* arm—the others are moving obstacles that are articulated, high-dimensional, and far from primitive shapes. Their joint states are usually available on the cell network, but exploiting this information for control-level safety is not trivial: a collision constraint must be re-evaluated at every control step, it involves the geometry of every link pair, and it must anticipate that the obstacle keeps moving while the ego arm reacts.

Control barrier functions (CBFs) [1], [2] enforce safety at the control level by filtering a nominal command through a quadratic program (QP), and have been applied to manipulators with analytic distance constraints [3], [4], [5], [6]. Analytic barriers, however, need an explicit environment model [7], [8], and for an articulated obstacle they reduce to a minimum over many pairwise distances that is non-smooth and expensive. Neural CBFs [9], [10], [11], [12], [13] learn the barrier from data and thus avoid hand-designed

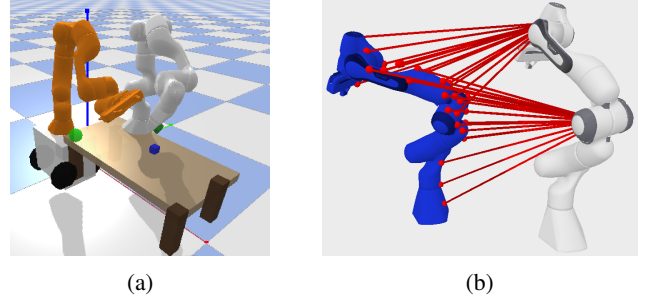


Fig. 1: **Setting.** (a) The ego arm (orange, on a mobile platform) executes a grasping task while a second arm (white) executes its own task in the same workspace and acts as a moving, articulated obstacle. (b) Ray-based observation: virtual range rays return first-hit points on the obstacle arm (blue), which are expressed in ego-link coordinate frames together with hit indicators.

geometry. In particular, *observation-based* neural CBFs [14], [15], [16], [17] take sensor-like inputs such as range rays or point clouds, so that the barrier can be defined on what the robot perceives rather than on a reconstructed model. A representative observation-based manipulator baseline [15] is formulated for static obstacles and uses numerical differentiation for the relevant derivative quantity. In a moving-obstacle setting, omitting obstacle-induced observation variation makes the corresponding CBF residual incomplete. Time-varying CBFs [18], [19], [20], [21] handle moving obstacles when an explicit obstacle model is available, but not for barriers learned on observations.

In this paper we make observation-based neural CBFs aware of obstacle motion. The setting is a shared workspace in which the ego arm avoids a second arm whose joint configuration q_e and joint velocity $\dot{q}_e$ are communicated (Fig. 1). The ego arm observes the obstacle through a ray-based observation $o = O(q, q_e)$ that depends on both its own configuration q and on q_e . Consequently, the time derivative of a barrier $h(q, o)$ contains an *obstacle-motion term* $\frac{\partial h}{\partial o} \frac{\partial O}{\partial q_e} \dot{q}_e$ in addition to the usual ego-motion term. Ray casting is not differentiable, so we learn a smooth surrogate $\hat{o} = \varphi_\theta(q, q_e)$ of the sensor model and use only its Jacobians in place of those of O . The barrier value is always evaluated on the real ray observation $o = O(q, q_e)$; the surrogate is used only to approximate the observation sensitivities needed in the complete CBF derivative during training and online filtering. Our contributions are:

- 1) an observation-based CBF formulation for moving

obstacles with communicated state, in which the CBF condition is written with the complete chain rule of $h(q, O(q, q_e))$, including the obstacle-motion term;

- 2) a learned differentiable observation model used only for the observation Jacobians, allowing the barrier itself to remain conditioned on real ray observations while the CBF derivative is evaluated without finite differencing the barrier output;
- 3) a verified simulation study over 1000 randomized trials with two 7-DoF Franka Panda arms, in which the proposed filter improves the collision-free rate from 64.0% to 82.0% and the task-success rate from 86.8% to 96.8% relative to a static-environment neural CBF adapted from [15]; an offline derivative diagnostic additionally compares automatic differentiation through the smooth surrogate with finite-difference estimates.

A video of the simulation experiments is submitted as supplementary material.

We state the assumptions and the limitations of the study explicitly (Sec. V): the obstacle state is assumed to be communicated, the evaluation is in simulation, and the learned barrier does not come with a formal guarantee.

II. PROBLEM FORMULATION

A. Ego Robot, Obstacle Robot, and Observation

The ego manipulator has configuration $q \in \mathcal{Q} \subseteq \mathbb{R}^n$ and control-affine kinematics

$$\dot{q} = f(q) + g(q)u, \quad u \in \mathcal{U} = \{u \in \mathbb{R}^m : |u_j| \leq \bar{u}_j\}, \quad (1)$$

with a box input set $\mathcal{U}$. In our experiments the arm is velocity-controlled, i.e., $f \equiv 0$ and $g \equiv I$, but the derivation does not rely on this.

The workspace is shared with an *obstacle manipulator* with configuration $q_e(t) \in \mathcal{Q}_e \subseteq \mathbb{R}^{n_e}$. The obstacle arm follows its own controller, which the ego arm can neither influence nor predict; the only information used about it is stated next.

Assumption 1. *At every control instant, the ego controller has access to the current configuration $q_e(t)$ and joint velocity $\dot{q}_e(t)$ of the obstacle manipulator, with $\dot{q}_e(t) \in \mathcal{V}_e$ for a known bounded set $\mathcal{V}_e$ (its joint-velocity limits).*

Assumption 1 matches a multi-robot cell in which every arm broadcasts its joint encoder readings; it does *not* require a model of the obstacle's controller or of its future trajectory. It provides the instantaneous obstacle information used by our formulation; no constant-velocity or other extrapolation of the obstacle motion is used; since the CBF condition is enforced at the control rate, the only implicit assumption is that $\dot{q}_e$ varies little within one control period. Scenarios in which q_e must be estimated from sensors (e.g., a human co-worker) are outside the scope of this paper, see Sec. V.

The ego arm perceives the obstacle through a *ray-based observation*. A virtual range sensor casts a fixed bundle of rays, giving R raw ray returns. For ray i we record the first intersection point $p_i \in \mathbb{R}^3$ with the obstacle arm and a hit

indicator $m_i \in \{0, 1\}$ (p_i is a fixed placeholder when $m_i = 0$). The raw returns are deterministically expressed in ego-link coordinate frames before being passed to the barrier encoder; for compactness, we denote the resulting fixed-size ray representation by

$$o = O(q, q_e) = \{(p_i, m_i)\}_{i=1}^R \in \mathcal{O} \subset \mathbb{R}^{R \times 4}. \quad (2)$$

Note that O depends on *both* arguments: the sensor pose and link-frame transformations depend on the ego configuration q , while the hit points also depend on the obstacle configuration q_e (Fig. 1b). In simulation, O is computed by ray casting against the obstacle meshes; on a physical cell it would be produced by range sensors.

B. Safety Specification

Collision status is determined by the pair $x = (q, q_e) \in \mathcal{X} := \mathcal{Q} \times \mathcal{Q}_e$. Let $d(x)$ be the minimum separation distance between the ego arm and the obstacle arm ($d(x) < 0$ under penetration), available offline from the simulator's collision checker but *not* used by the controller. Let $C_{\text{self}}(q) \in \{0, 1\}$ denote the ego-arm self-collision indicator. The implemented check flags configurations with a checked pair of non-adjacent links within 0.01 m of each other. For a margin $r_s > 0$ we partition $\mathcal{X}$ into

$$\begin{aligned} \mathcal{X}_u &= \{x \in \mathcal{X} : d(x) < 0 \text{ or } C_{\text{self}}(q) = 1\}, \\ \mathcal{X}_s &= \{x \in \mathcal{X} : d(x) \geq r_s, C_{\text{self}}(q) = 0\}, \\ \mathcal{X}_\partial &= \mathcal{X} \setminus (\mathcal{X}_u \cup \mathcal{X}_s). \end{aligned} \quad (3)$$

These are *geometric labels* of configuration pairs. Configurations flagged by the self-collision check are included in $\mathcal{X}_u$ and excluded from $\mathcal{X}_s$, so self-collision is explicitly accounted for in the training labels. The barrier introduced next is a function of (q, o) and receives neither d nor C_{self} as an explicit input.

We use the sublevel-set sign convention of [15]: negative values of h are safe. (The standard superlevel form is recovered with $b = -h$.)

Definition 1 (Observation-based CBF for a moving obstacle). *Let $h : \mathcal{Q} \times \mathcal{O} \rightarrow \mathbb{R}$ be continuously differentiable and write $\tilde{h}(q, q_e) := h(q, O(q, q_e))$. Given $\alpha > 0$, $\gamma > 0$, and $\epsilon \geq 0$, h is an observation-based CBF for (1)–(2) if*

- i) $\tilde{h}(x) \leq -\gamma$ for all $x \in \mathcal{X}_s$;
- ii) $\tilde{h}(x) \geq \gamma$ for all $x \in \mathcal{X}_u$;
- iii) for all $x = (q, q_e)$ with $\tilde{h}(x) \leq 0$ and all $\dot{q}_e \in \mathcal{V}_e$,

$$\inf_{u \in \mathcal{U}} \left[\frac{\partial \tilde{h}}{\partial q} (f(q) + g(q)u) + \frac{\partial \tilde{h}}{\partial q_e} \dot{q}_e \right] \leq -\alpha \tilde{h}(x) - \epsilon. \quad (4)$$

By the chain rule, the two derivative terms in (4) are

$$\frac{\partial \tilde{h}}{\partial q} = \frac{\partial h}{\partial q} + \frac{\partial h}{\partial o} \frac{\partial O}{\partial q}, \quad \frac{\partial \tilde{h}}{\partial q_e} = \frac{\partial h}{\partial o} \frac{\partial O}{\partial q_e}, \quad (5)$$

i.e., the ego motion enters both directly through q and through the sensor pose, and the obstacle motion enters only through the observation. The second term of (4) plays the role of the explicit time derivative in time-varying CBFs [18]; here it is induced by $\dot{q}_e$ through the sensor model.

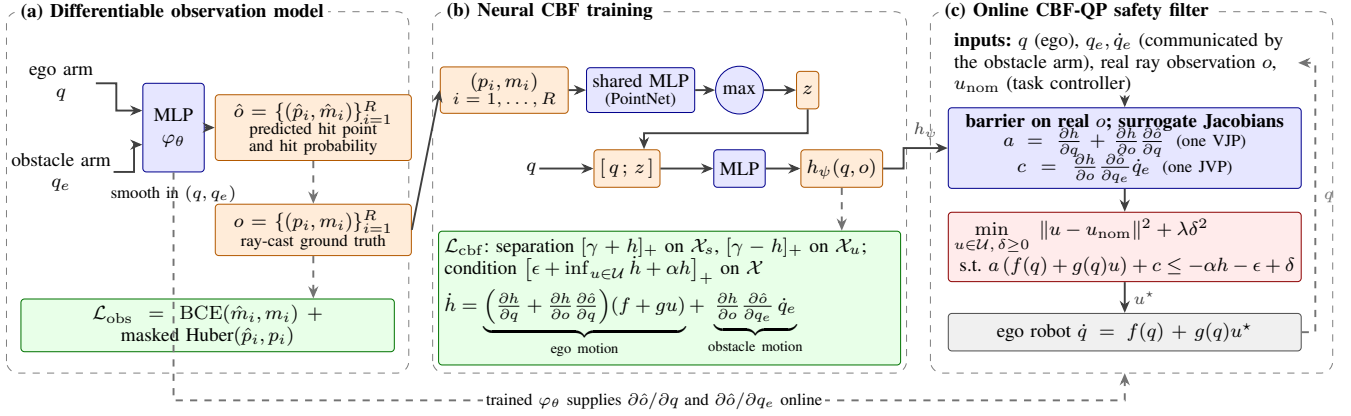


Fig. 2: **Overview.** (a) A multilayer perceptron φ_θ is trained to approximate the ray-cast observation map. (b) The neural CBF $h_\psi(q, o)$ is evaluated on the *real* ray observation $o = O(q, q_e)$, while φ_θ is used only to approximate the observation Jacobians in the CBF derivative. (c) At run time, the real observation supplies the barrier input, while the surrogate supplies the observation sensitivities used by the CBF-QP.

Define the induced set $\mathcal{C} = \{x \in \mathcal{X} : \tilde{h}(x) \leq 0\}$. Conditions i)–ii) give $\mathcal{X}_s \subseteq \mathcal{C}$ and $\mathcal{C} \cap \mathcal{X}_u = \emptyset$; states in the margin band $\mathcal{X}_\partial$ may or may not belong to $\mathcal{C}$. For an exact continuously differentiable observation map and barrier, condition iii) is the standard CBF condition [1] applied to the time-varying function $t \mapsto \tilde{h}(q(t), q_e(t))$; if the unrelaxed inequality is satisfied at every instant, $\mathcal{C}$ is forward invariant and trajectories starting in $\mathcal{X}_s \subseteq \mathcal{C}$ avoid $\mathcal{X}_u$. Importantly, iii) is what excludes from $\mathcal{C}$ those configurations that are currently separated from the obstacle but cannot remain so under the input bounds and the obstacle velocity; this is why a classifier of $\mathcal{X}_s$ versus $\mathcal{X}_u$ alone is not a barrier and why the derivative terms in (5) must be enforced during training.

Problem 1. *Given the ego arm (1), the observation (2), Assumption 1, and a nominal task controller u_{nom} , synthesize an observation-based CBF h satisfying Definition 1 and an online controller $u = \pi(q, o, q_e, \dot{q}_e)$ that keeps $x(t) \in \mathcal{C}$ for all $t \geq 0$ whenever $x(0) \in \mathcal{C}$, while deviating minimally from u_{nom} .*

Since h will be represented by a neural network, conditions i)–iii) can only be enforced on samples and hold approximately; the resulting closed-loop safety is therefore verified empirically in Sec. IV, and the residual violations are reported rather than hidden.

III. METHOD

Fig. 2 summarizes the approach. Offline, we (a) learn a differentiable surrogate $\hat{o} = \varphi_\theta(q, q_e)$ of the ray-casting sensor model and (b) train a neural CBF $h_\psi(q, o)$ directly on the real ray observation $o = O(q, q_e)$. The surrogate is used only to approximate the observation Jacobians in (5). Online, (c) a CBF-QP filters u_{nom} using the current real ray observation for the barrier value and φ_θ only for the observation sensitivities.

A. Differentiable Surrogate of the Ray-Casting Sensor

The map O in (2) is only piecewise smooth: a ray’s hit point jumps when the ray slides off one link onto another or misses the obstacle, and the hit indicator is binary. The Jacobians in (5) are therefore undefined or uninformative on a set that is visited frequently in practice. We replace O by a learned smooth surrogate

$$\hat{o} = \varphi_\theta(q, q_e) = \{(\hat{p}_i, \hat{m}_i)\}_{i=1}^R, \quad \hat{m}_i \in (0, 1), \quad (6)$$

where φ_θ is a fully connected network with SiLU (sigmoid linear unit) activations, taking the concatenation (q, q_e) as input and producing R predicted hit points and hit probabilities. Training data $\mathcal{D}_{\text{obs}} = \{(q^{(k)}, q_e^{(k)}, o^{(k)})\}_{k=1}^{N_{\text{obs}}}$ are generated by sampling configuration pairs and ray casting in the simulator. The loss combines a per-ray binary cross-entropy (BCE) hit classification and a hit-masked regression of the hit points,

$$\mathcal{L}_{\text{obs}} = \frac{1}{R} \sum_{i=1}^R \text{BCE}(\hat{m}_i, m_i) + \frac{\sum_{i=1}^R m_i \rho_\delta(\|\hat{p}_i - p_i\|_2)}{\sum_{i=1}^R m_i + \varepsilon_0}, \quad (7)$$

with the Huber penalty ρ_δ and a small $\varepsilon_0 > 0$; the mask m_i restricts the regression to rays that actually hit the obstacle.

Relation to geometry-based alternatives. The surrogate (6) predicts the observation associated with a configuration pair, but the predicted observation is not used as the barrier input in our experiments. Instead, h_ψ receives the real ray observation, and φ_θ is used only to provide smooth configuration sensitivities that approximate the Jacobians of the non-differentiable ray-casting map. Analytic distance-based CBFs and learned configuration-space distance fields [22], [23], [24] are valid alternatives when an accurate robot/obstacle geometry model is available. We do not compare against those alternatives in the present study (Sec. V).

Dependence on obstacle motion. The surrogate is a function of configurations only; the obstacle velocity enters the method exclusively through the linear term $\frac{\partial \hat{o}}{\partial q_e} \dot{q}_e$. Its validity

therefore depends on the coverage of $\mathcal{Q} \times \mathcal{Q}_e$ by $\mathcal{D}_{\text{obs}}$, not on the particular obstacle trajectories seen at training time. Fast obstacle motion remains harder for the *controller* (bounded $\mathcal{U}$), but does not by itself drive φ_θ out of distribution.

B. Neural CBF: Architecture and Training

Architecture. Following Fig. 2(b), $h_\psi(q, o)$ applies a shared multilayer perceptron to each ray feature (p_i, m_i) , max-pools the per-ray features into a fixed-size code z (a PointNet-style encoder [25]), concatenates z with q , and maps the result through a second multilayer perceptron to a scalar. Max pooling lets h respond to the closest hit points, which dominate collision risk, and keeps the network size independent of R .

Data. Similar to [15], the training set $\mathcal{D}_{\text{cbf}}$ is generated from two sources: (i) closed-loop rollouts of the nominal controller from random start and goal configurations while the obstacle arm executes randomized tasks, which concentrates samples in the region the controller actually visits, and (ii) uniform samples of (q, q_e) in $\mathcal{Q} \times \mathcal{Q}_e$ to improve coverage. Each sample $(q, q_e, \dot{q}_e)$ is labeled safe, unsafe, or margin according to (3) with the simulator's collision checker; $\dot{q}_e$ is taken from the rollout or sampled in $\mathcal{V}_e$.

Loss. With N_s , N_u , and N the numbers of samples in $\mathcal{X}_s$, $\mathcal{X}_u$, and $\mathcal{X}$, and $[\cdot]_+ = \max(0, \cdot)$, we minimize

$$\mathcal{L}_{\text{cbf}} = \frac{\alpha_1}{N_s} \sum_{x \in \mathcal{X}_s} [\gamma + h]_+ + \frac{\alpha_2}{N_u} \sum_{x \in \mathcal{X}_u} [\gamma - h]_+ + \frac{\alpha_3}{N} \sum_{x \in \mathcal{X}} \left[\epsilon + \inf_{u \in \mathcal{U}} \dot{h}(x, u, \dot{q}_e) + \alpha h \right]_+, \quad (8)$$

where h is short for $h_\psi(q, o)$ with the real observation $o = O(q, q_e)$, and

$$\begin{aligned} \dot{h}(x, u, \dot{q}_e) = & \underbrace{\left(\frac{\partial h}{\partial q} + \frac{\partial h}{\partial o} \frac{\partial \hat{o}}{\partial q} \right) (f + gu)}_{\text{ego motion}} \\ & + \underbrace{\frac{\partial h}{\partial o} \frac{\partial \hat{o}}{\partial q_e} \dot{q}_e}_{\text{obstacle motion}}, \quad h = h_\psi(q, o), \quad o = O(q, q_e). \end{aligned} \quad (9)$$

The first two terms of (8) enforce i)–ii) of Definition 1 with margin γ ; the third enforces iii) with margin ϵ . Because $\dot{h}$ is affine in u and $\mathcal{U}$ is a box, the inner infimum is available in closed form: writing $a = \frac{\partial h}{\partial q} + \frac{\partial h}{\partial o} \frac{\partial \hat{o}}{\partial q}$,

$$\inf_{u \in \mathcal{U}} \dot{h} = af(q) - \sum_{j=1}^m \bar{u}_j |(ag(q))_j| + \frac{\partial h}{\partial o} \frac{\partial \hat{o}}{\partial q_e} \dot{q}_e. \quad (10)$$

The partial derivatives $\partial h / \partial q$ and $\partial h / \partial o$ are evaluated at the real observation $o = O(q, q_e)$. The observation Jacobians $\partial O / \partial q$ and $\partial O / \partial q_e$ are approximated by $\partial \hat{o} / \partial q$ and $\partial \hat{o} / \partial q_e$ from the frozen surrogate φ_θ . The resulting CBF loss is minimized with standard first-order optimizers, while θ remains fixed during barrier training.

Remark 1 (Relation to static observation-based neural CBFs). *The neural CBF of [15], which we use as the*

baseline, is trained with the ego-motion term only, i.e., it assumes $\dot{o}$ to be negligible, and it evaluates $\frac{\partial h}{\partial q}$ by finite differences of h along perturbed configurations. Both choices are natural for static scenes but discard the obstacle-motion term of (9), which is exactly the part of $\dot{h}$ that a moving obstacle contributes. In addition, finite differencing a network output trades truncation error (large step) against noise amplification (small step); automatic differentiation of the surrogate sensitivities avoids a finite-difference step-size parameter. Sec. IV-D compares the resulting derivative estimates on a shared set of offline states.

C. Online Safety Filter

Given the trained h_ψ and φ_θ , the ego arm is controlled at every step by the CBF-QP

$$\begin{aligned} \min_{u \in \mathcal{U}, \delta \geq 0} & \|u - u_{\text{nom}}\|_2^2 + \lambda \delta^2 \\ \text{s.t.} & a(f(q) + g(q)u) + c \leq -\alpha h - \epsilon + \delta, \end{aligned} \quad (11)$$

whose minimizer (u^*, δ^*) gives the applied input u^* . At every control step, the barrier value is evaluated on the current real ray observation,

$$h = h_\psi(q, o), \quad o = O(q, q_e).$$

Define

$$a = \frac{\partial h}{\partial q} + \frac{\partial h}{\partial o} \frac{\partial \hat{o}}{\partial q}, \quad c = \frac{\partial h}{\partial o} \frac{\partial \hat{o}}{\partial q_e} \dot{q}_e,$$

where the derivatives $\partial h / \partial q$ and $\partial h / \partial o$ are evaluated at the measured ray observation o , while $\partial \hat{o} / \partial q$ and $\partial \hat{o} / \partial q_e$ are obtained from the differentiable surrogate φ_θ . Thus the surrogate never replaces the real observation in the barrier input; it only supplies the observation sensitivities required by the chain rule. In implementation, the gradients of $h_\psi(q, o)$ with respect to (q, o) are obtained by reverse-mode automatic differentiation. The term $(\partial h / \partial o)(\partial \hat{o} / \partial q)$ is a vector–Jacobian product through φ_θ , and $(\partial \hat{o} / \partial q_e) \dot{q}_e$ is a Jacobian–vector product through φ_θ . The QP has $m + 1$ variables, one linear inequality, and box bounds; the differentiable soft-QP implementation uses SCS through CVXPYLayers. The nonnegative slack δ keeps (11) feasible, while the quadratic penalty discourages relaxation of the CBF condition. Positive δ^* quantifies violation of the unrelaxed certificate by the selected input and can be reported as a separate online diagnostic; it does not by itself establish that no admissible input satisfies the unrelaxed constraint.

Remark 2 (Several obstacle arms). *With K obstacle arms sharing the workspace, the observation concatenates the per-obstacle ray returns, the surrogate takes $(q, q_{e,1}, \dots, q_{e,K})$ as input, and the obstacle-motion term becomes $\sum_k \frac{\partial h}{\partial o} \frac{\partial \hat{o}}{\partial q_{e,k}} \dot{q}_{e,k}$. Two caveats prevent us from claiming this extension without evidence: a ray's hit may switch from one obstacle to another, which introduces additional discontinuities that the surrogate must be trained across, and barriers composed as minima over obstacles are non-smooth [26]. All experiments in this paper use a single obstacle arm.*

IV. EXPERIMENTAL EVALUATION

A. Setup

Simulation. All experiments run in PyBullet [27] with two 7-DoF Franka Panda arms (Fig. 1a). The ego arm is mounted on a mobile platform whose base is held fixed during the task. The controller operates at 30 Hz, with a PyBullet simulation frequency of 120 Hz. The ego-arm joint-velocity limits are $\bar{u}_j = 2.175$ rad/s for joints 1–4 and $\bar{u}_j = 2.61$ rad/s for joints 5–7. The obstacle arm executes randomized motions with joint velocities bounded by the same joint-specific limits; its initial configuration and motion targets are randomized across trials to produce diverse relative motions. The experimental observation backend uses a single virtual range sensor whose origin is attached to the middle link of the ego arm (`panda_link4`). At every observation step, the sensor casts $R = 2,048$ rays with a maximum range of 5 m over an 85° cone directed toward the obstacle arm. The resulting point cloud is transformed into seven ego-link coordinate frames and sampled to 256 points per frame, giving 1,792 point representations as input to the barrier observation encoder. This deterministic preprocessing is applied consistently to the real ray observation o that is fed to the learned barrier.

Task and nominal controller. The ego arm must bring its end-effector to a randomized grasp pose within a horizon of 10 s. The implemented nominal controller is joint-space feedback with input saturation, $u_{\text{nom}} = \text{clip}_{\mathcal{U}}[-K(q - q_{\text{goal}})]$. The gain K is obtained by discrete-time LQR with identity state and input cost matrices, using the configured controller period. In the grasping scene, inverse kinematics provides the joint-space target q_{goal} . This nominal feedback uses the joint configuration and target and does not depend on obstacle observations.

Networks and training. The available φ_θ checkpoints use an MLP with three hidden layers of 256 units each and SiLU activations, trained on $N_{\text{obs}} = 7,920$ configuration pairs. The implemented barrier h_ψ appends a one-hot ego-link-frame identifier to each point and applies a shared per-ray MLP with layer widths 64, 128, and 512 and LeakyReLU activations with negative slope 0.1. Max pooling over the rays produces a 512-dimensional feature for each ego-link frame. A per-frame MLP with an intermediate width of 256 projects these features to 64 dimensions, which are then concatenated and fused by an MLP with intermediate widths 512 and 256 into a 32-dimensional observation code. The scalar barrier head has three hidden layers of 48 units each with LeakyReLU activations of negative slope 0.1. The CBF data-generation configuration uses 120 episodes, each with 40 rollouts of 35 states and 200 uniformly sampled states. This specifies 192,000 labeled samples, of which 168,000 (87.5%) come from rollouts. A random 90/10 training–validation split gives 172,800 training samples and 19,200 validation samples for a fully generated dataset. The barrier is trained with $r_s = 0.05$ m, $\gamma = 0.1$, $\epsilon = 0$, $\alpha = 1$, $(\alpha_1, \alpha_2, \alpha_3) = (20, 20, 2)$. The quadratic slack penalty in (11) uses $\lambda = 5000$. Training is performed on an NVIDIA

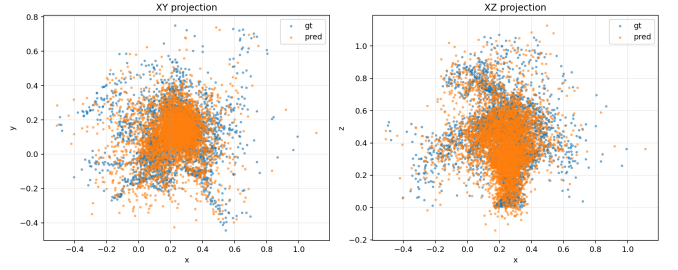


Fig. 3: **Surrogate sensor model.** Ground-truth ray-cast hit points (blue) and predictions of φ_θ (orange) on evaluated configuration pairs on held-out configuration pairs, projected on the xy and xz planes (units: m).

GeForce RTX 5090 GPU.

Baseline. We compare against a static-environment neural CBF adapted from [15] (S-NCBF). To isolate the contribution, the baseline uses the *same* observation, network architecture, training data, nominal controller, and QP as OD-NCBF; it differs only in the two respects of Remark 1: the obstacle-motion term is omitted in training and online, and $\frac{\partial h}{\partial q}$ is approximated by forward finite differences with joint perturbation $s = 1/30$ rad, using $[h(q + se_j, o(q + se_j)) - h(q, o(q))]/s$. Other neural or analytic baselines (e.g., distance-based CBFs) are not included, see Sec. V.

Protocol and metrics. Each method is evaluated on the same 1000 randomized episodes (initial configurations of both arms, obstacle task, and grasp target). We report the *success rate*, the fraction of episodes in which the grasp pose is reached within the horizon, and the *safety rate*, the fraction of episodes without any contact between the arms as detected by the simulator at every step. An episode is marked unsafe at the first detected contact and is terminated immediately thereafter. Accordingly, rollout trajectories are truncated at the first collision. For each marginal rate we give a 95% Wilson confidence interval [28]. Because the two methods are evaluated on the same 1000 randomized episodes, comparisons of the paired binary outcomes use a two-sided McNemar test.

B. Accuracy of the Surrogate Sensor Model

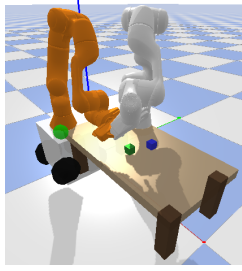
Fig. 3 compares the hit points predicted by φ_θ with the ray-cast ground truth on held-out configuration pairs. The predictions follow the ground-truth distribution closely, including the tails corresponding to rare hits on the distal links; the mean hit-point error is 0.0532 m and the hit/miss classification accuracy is 97.58%. This level of accuracy is sufficient for the surrogate’s role, which is to supply smooth Jacobians rather than to replace the sensor.

C. Closed-Loop Comparison

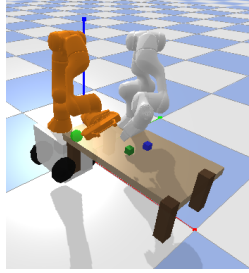
Table I reports the verified 1000-episode comparison. OD-NCBF raises the safety rate from 0.640 to 0.820 and the success rate from 0.868 to 0.968; paired McNemar tests on the episode-level binary outcomes give $p < 0.001$ for both metrics. The higher success rate accompanies the higher collision-free rate; in failed baseline rollouts, collisions often

TABLE I: **Verified closed-loop results.** Results over the same 1000 randomized episodes (S-NCBF: static-environment neural CBF with finite-difference derivatives; OD-NCBF: proposed). Brackets: 95% Wilson confidence intervals. Paired McNemar tests on the episode-level binary outcomes give $p < 0.001$ for both Success and Safety.

Metric	S-NCBF [15]	OD-NCBF (ours)	Δ
Success rate	0.868 [0.846, 0.888]	0.968 [0.955, 0.977]	+0.100
Safety rate	0.640 [0.610, 0.669]	0.820 [0.795, 0.843]	+0.180



(a) S-NCBF (baseline)



(b) OD-NCBF (ours)

Fig. 4: **Snapshots** from the same episode at the instant of closest approach. The baseline collides with the obstacle arm (white); the proposed filter keeps a clear separation.

deflect or block the ego arm so that the grasp pose is not reached in time.

Fig. 4 and Fig. 5 show a representative episode. Under the baseline, the barrier value chatters between its safe level and values above zero for about two seconds while the obstacle arm sweeps through the ego workspace (Fig. 5a); this is the signature of a constraint whose coefficients are inconsistent from step to step: with the obstacle-motion term missing and a finite-difference gradient, the QP alternately activates and releases the constraint, producing jittery commands, and the separation distance eventually becomes negative, i.e., the arms penetrate (Fig. 5c). Under OD-NCBF, the barrier stays at its safe level with small, smooth variations (Fig. 5b); the ego arm yields as the obstacle approaches, the separation distance stays positive with a minimum of about 3.5 cm (Fig. 5d), and the task is completed once the obstacle has passed.

Remaining failures. OD-NCBF still collides in 18% of the episodes, a rate that is far from what deployment would require. Inspecting the failed episodes we identify three causes. First, in some initial configurations the obstacle arm sweeps through the ego arm’s current position faster than the ego arm can retreat under its velocity bounds; in such cases the unrelaxed CBF constraint may be infeasible under the input bounds. This concerns certificate feasibility and does not establish that every admissible control leads to collision; positive slack alone also does not prove infeasibility. Second, the learned h is imprecise in the margin band $\mathcal{X}_\delta$, where the hinge losses in (8) are not driven to zero. Third, surrogate

errors on rare ray configurations perturb the constraint coefficients. We return to these points in Sec. V.

D. Derivative Diagnostic: Surrogate-Jacobian Autodiff versus Finite Differences

The comparison above changes both the obstacle-motion term and the derivative computation. To isolate the latter, we evaluate the same trained barrier h_ψ on the same of-line states and under the same control inputs, retaining the obstacle-motion term in both variants. We compare the surrogate-Jacobian autodiff estimate, which evaluates h_ψ on the real observation and uses φ_θ only for the observation Jacobians, with the forward finite-difference estimate $\dot{h}_{\text{FD}} = [h(x_{k+1}) - h(x_k)]/\Delta t$, where $\Delta t = 1/30$ s is the controller period. For each evaluated state, we report the post-hoc CBF violation residual $\rho = \max\{0, \dot{h} + \alpha h\}$.

Fig. 6 reports the results. With the surrogate-Jacobian autodiff estimate, the mean residual decreases from about 0.22 to 0.06 and its 95th percentile from about 1.7 to 0.4 (Fig. 6a); the fraction of offline states with zero residual increases from about 0.83 to 0.89 (Fig. 6b). Because the barrier, evaluated states, and control inputs are shared, these differences arise from the derivative estimate. The finite-difference estimate depends on the perturbation step. Fig. 6c shows that its standard deviation across the evaluated states grows by roughly an order of magnitude when ε is reduced from 10^{-2} to 10^{-3} . Because this statistic is computed across sampled states rather than repeated noisy measurements of a fixed state, we interpret panel (c) as step-size sensitivity rather than as an isolated measurement-noise experiment. The surrogate-Jacobian autodiff estimate has no finite-difference step-size parameter and therefore avoids this tuning trade-off.

V. LIMITATIONS

We list the limitations of this study explicitly. *Communicated obstacle state.* Assumption 1 restricts the method to cells in which the obstacle arm shares its joint state. A human co-worker or an unmodeled agent would require estimating q_e and $\dot{q}_e$ from sensors; estimation errors then enter the obstacle-motion term, and a measurement-robust treatment of the CBF condition [29] would be needed. We have not evaluated this. *Simulation only.* All results are obtained in a noise-free simulator with virtual range sensors rigidly attached to the ego links. A physical cell would involve sensor noise, occlusions, latency, and imperfect velocity tracking, none of which is modeled here; Sec. IV-D studies finite-difference step-size sensitivity rather than physical sensor noise. Hardware validation is the most important next step. *Single obstacle arm.* The extension to several moving arms is formulated but not evaluated; ray re-assignment between obstacles introduces discontinuities that we have not studied. *No formal guarantee.* h_ψ is trained to satisfy Definition 1 on samples; the verified closed-loop evaluation still contains collisions, and the slack in (11) relaxes the certificate when the constraint cannot be satisfied without relaxation under the selected solution and input bounds. Verification of the learned barrier and a fallback strategy when $\delta^* > 0$ (e.g.,

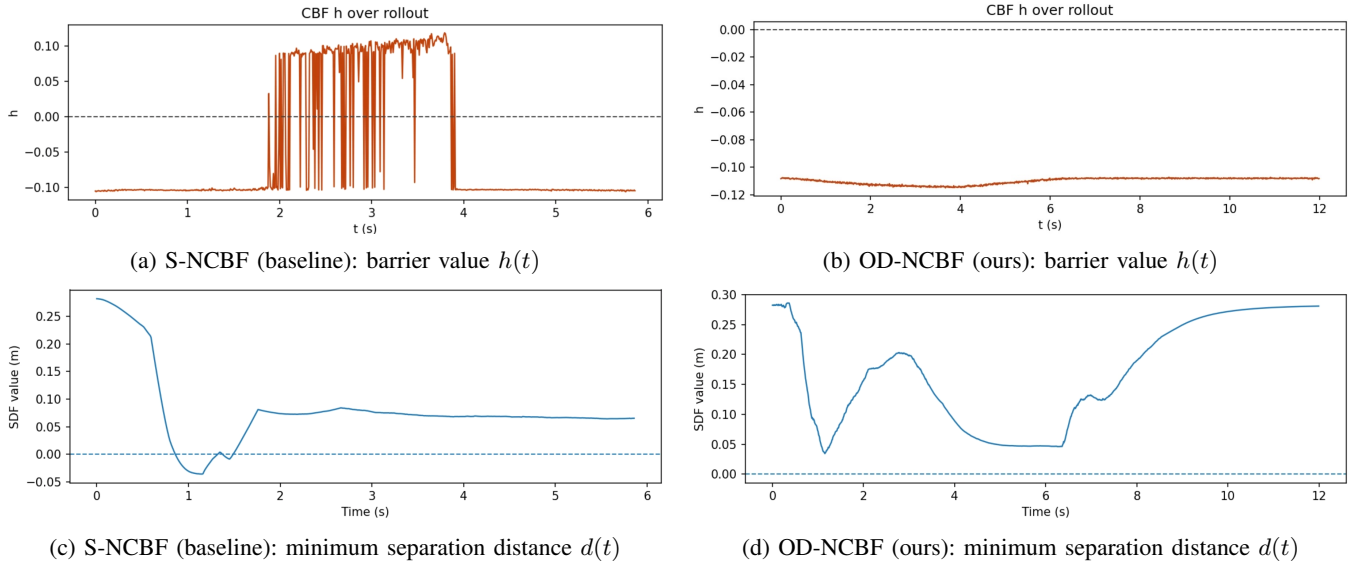


Fig. 5: **Representative episode.** Left column: baseline; right column: proposed method. Top: barrier value along the rollout (dashed line: $h = 0$, values below are certified safe). Bottom: minimum separation distance between the arms measured by the simulator (labeled SDF in the plots; dashed line: contact). The baseline trajectory is truncated at the first detected collision, whereas OD-NCBF remains collision-free throughout the episode.

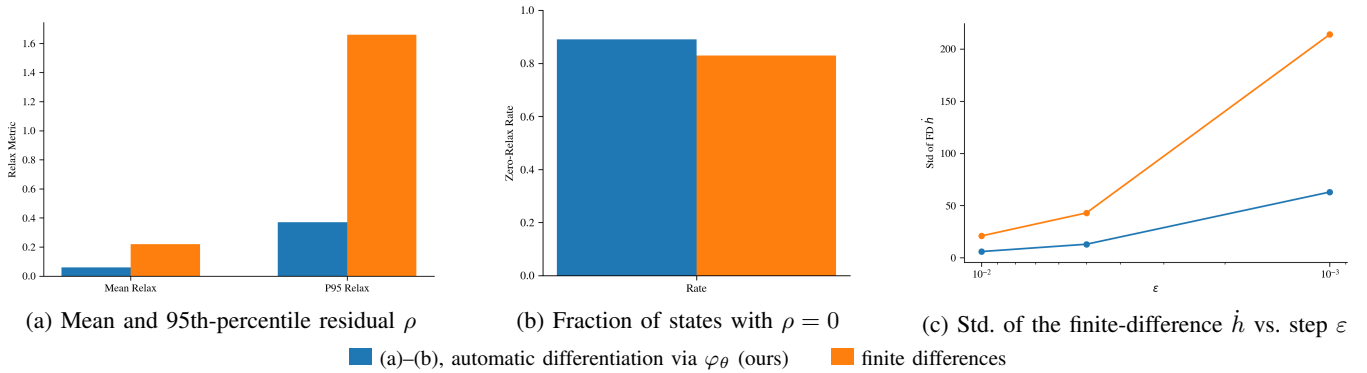


Fig. 6: **Offline derivative diagnostic** using the same trained barrier, offline states, and control inputs. (a)–(b) Statistics of the post-hoc CBF violation residual over five random-seed evaluation runs: automatic differentiation yields an approximately $3.5\times$ smaller mean residual (0.06 versus 0.22) and an approximately $4.5\times$ smaller 95th-percentile residual (0.37 versus 1.66), while yielding zero residual in a larger fraction of offline states (0.89 versus 0.83). (c) Standard deviation of the finite-difference estimate of $\dot{h}$ as a function of the step ϵ , for model-generated observations (E1, blue) and raw observations (E2, orange). The estimate becomes increasingly variable as ϵ decreases, whereas the surrogate-Jacobian autodiff estimate is independent of ϵ .

stopping or retreating) are left for future work. *Baselines.* We compare against a single, closely matched neural baseline in order to isolate the effect of the obstacle-motion term and of the derivative computation. Analytic distance-based CBFs, which are applicable here because q_e is known, and learned distance fields were not included and might be competitive.

VI. CONCLUSION

We presented an observation-based neural CBF for a manipulator that shares its workspace with a second, moving manipulator whose joint state is communicated. Writing the CBF condition with the complete time derivative of $h(q, O(q, q_e))$ exposes an obstacle-motion term that static

observation-based formulations neglect; a learned smooth surrogate of the ray-casting sensor supplies the observation Jacobians needed to train and evaluate this term, while the barrier itself is conditioned on the real ray observation during both training and online control. In the verified 1000-trial simulation comparison, OD-NCBF increases the collision-free rate from 64.0% to 82.0% and the task-success rate from 86.8% to 96.8% relative to S-NCBF, with both differences significant under paired McNemar tests ($p < 0.001$). The offline derivative diagnostic further shows smaller post-hoc CBF residuals when the observation sensitivities are obtained by automatic differentiation through the smooth surrogate than when finite-difference estimates are used on the

evaluated states. The remaining collisions, communicated-obstacle-state assumption, simulation-only evaluation, and lack of comparison to analytic geometry-based alternatives delimit the scope of the current results; hardware validation and robustness to estimated obstacle states are important next steps.

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